The Visual System

Topographic maps

Computational Neuroscience

University of Bristol

mrule

Learning outcomes:

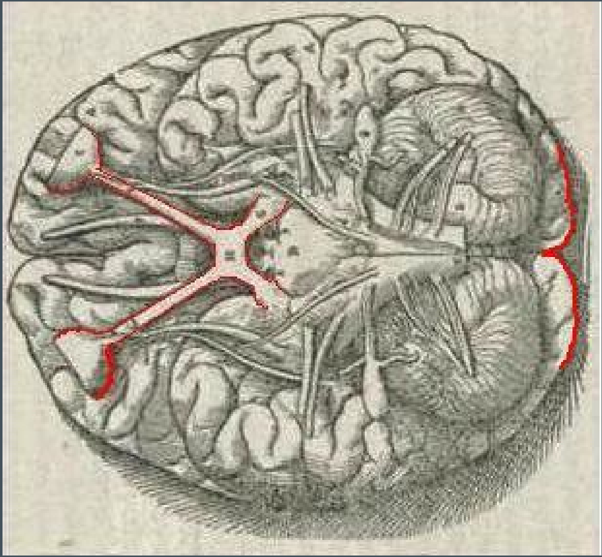
- How does the retina map onto V1?
- What is a topographic map (in the brain)?
- What is a Kohonen map?

Retinotopy & Topographic Maps

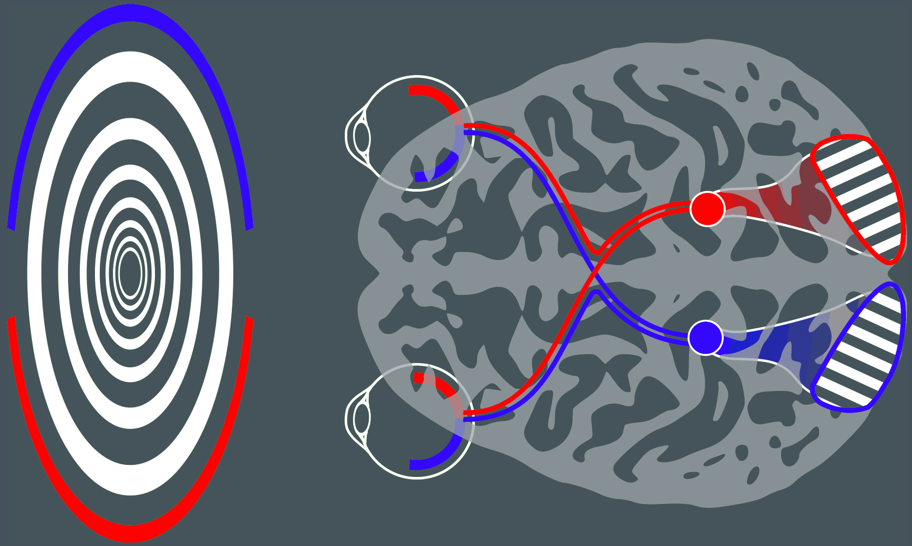
In development, chemical signals allow signals from the retina to preserve the 2D geometry of visual input in primary visual cortex (V1).

This is known as **retinotopy**.

The retinotopic map in V1 can be approximated as a log-polar transformation.



Andreas Vesalius (1543) De Humani Corporis Fabrica (Wikimedia)



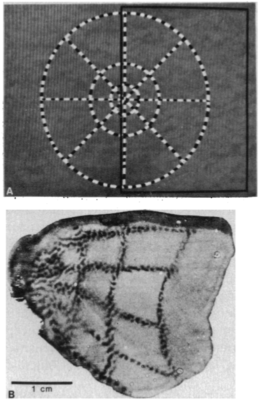
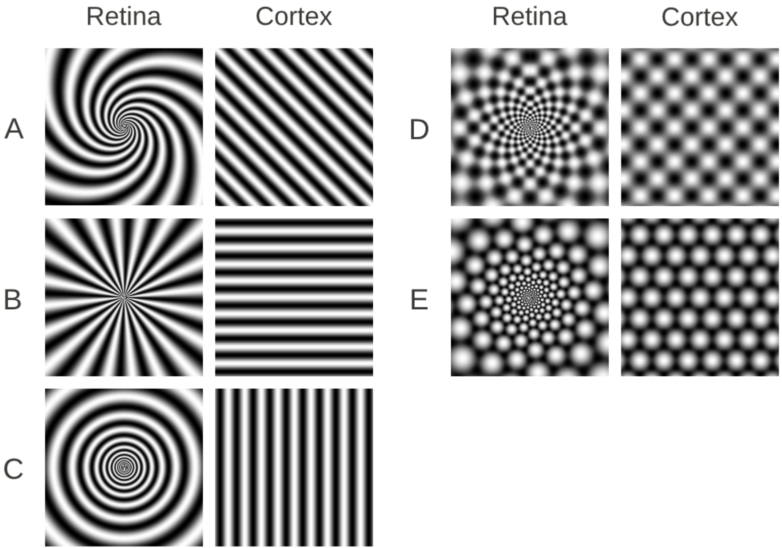


Fig. 1. (A) One of the visual stimuli used. The solid black rectangle encloses that portion of the visual stimulus that stimulated the region of striate cortex shown in (B). (B) Pattern of brain activation produced by the visual stimulus shown in (A), as revealed by 2DG. This is an autoradiograph from a single flat-mounted tissue section (mostly from layers 4B and 4C). About half of the total surface area of the macaque striate cortex can be seen.

Tootell (1982)

cadherin/netrin/ephrin-wnt/slit-robo/reelin/semaphorin ***not*** on exam

retinotopy: nearby regions respond to similar parts of receptive field



Rule, Stoffregen, Ermentrout (2011)

fig. 2

fig 3

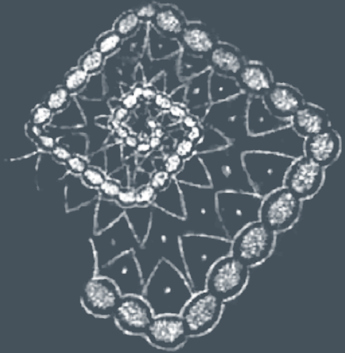


fig 4

Jan Purkinje (1819)

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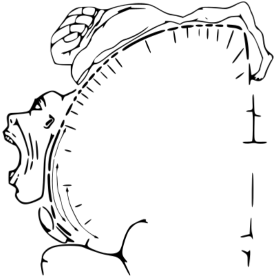
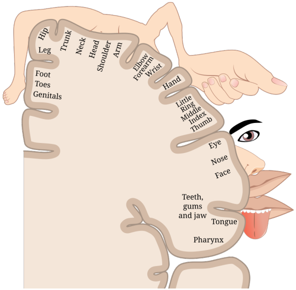
Pause

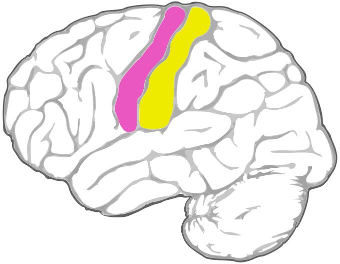
Topographic Maps more generally: Nearby neurons in cortex demonstrate similar functional properties



Motor homunculus, Penfield and Rasmussen (1950)



Sensory homunculus; Anatomy & Physiology, Connexions



Topographic maps

V1 in most mammals has multiple superimposed maps:

- Orientation preference
- Ocular dominance
- Motion direction preference
- Spatial frequency sensitivity
- Color sensitivity

Auditory cortex

- Tonotopic map

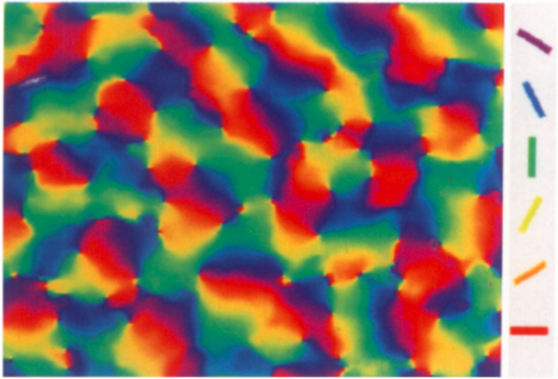
Somatosensory cortex

- Somatotopic map, sensory homunculus

Motor cortex cortex

- Motor homunculus

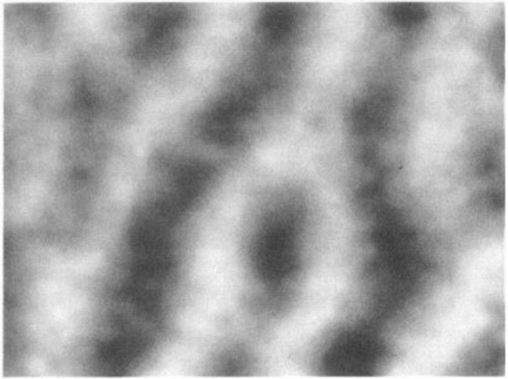
Orientation preference maps



Blasdel, J Neurosci (1992)

Orientation preference patches at the surface of macaque V1.

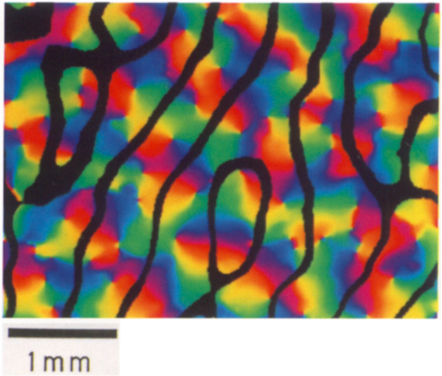
Ocular dominance maps



Blasdel, J Neurosci (1992)

Ocular dominance patches at the surface of macaque V1.

Interaction between OR and OD maps



Blasdel, J Neurosci (1992)

Pause

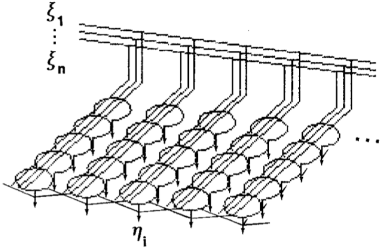
Kohonen / self-organising maps

Models of V1 development

- Various models have been proposed to explain aspects of visual cortex development.
- Ocular dominance columns classic example (Kenneth Miller). Requires symmetry-breaking.
- Kohonen map (a.k.a. self-organising map) can generically explain topographic map formation.

The Kohonen map algorithm for learning self-organised maps

Neuronal activation $\boldsymbol{y} = \mathbf{W}\boldsymbol{x}$ a linear function of inputs $\boldsymbol{x}$



Kohonen, Biol Cybern (1982)

1. Present input $\boldsymbol{x}$
2. Find the most active neuron in the array, $\max_i(y_i)$
3. Update weights of most active neuron + its nearest neighbours as

$$w_{i;t+1} \leftarrow \frac{w_{i;t} + \eta_t x_t}{\|w_{i;t} + \eta_t x_t\|},$$

   where learning rate η_t decreases over time.

4. Repeat from step 1. This results in the weights of neighbouring units becoming more similar with training.

Kohonen map

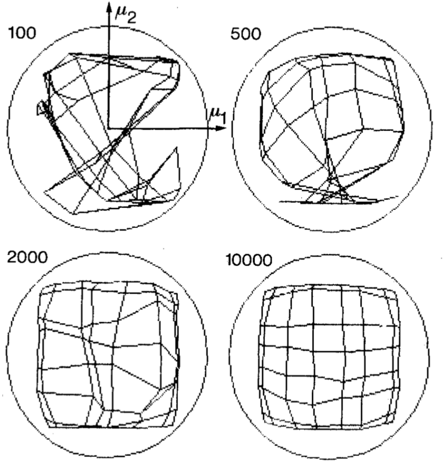


Fig. 4. Distribution of the weight vectors $m_i(t)$ at different times. The number of training steps is shown above the distribution. Interaction between nearest neighbours only

Kohonen, Biol Cybern (1982)

https://githubtocolab.com/engmaths/SEMT30003_2025/blob/main/Labs/lab7/self_organizing_map.ipynb *end*

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